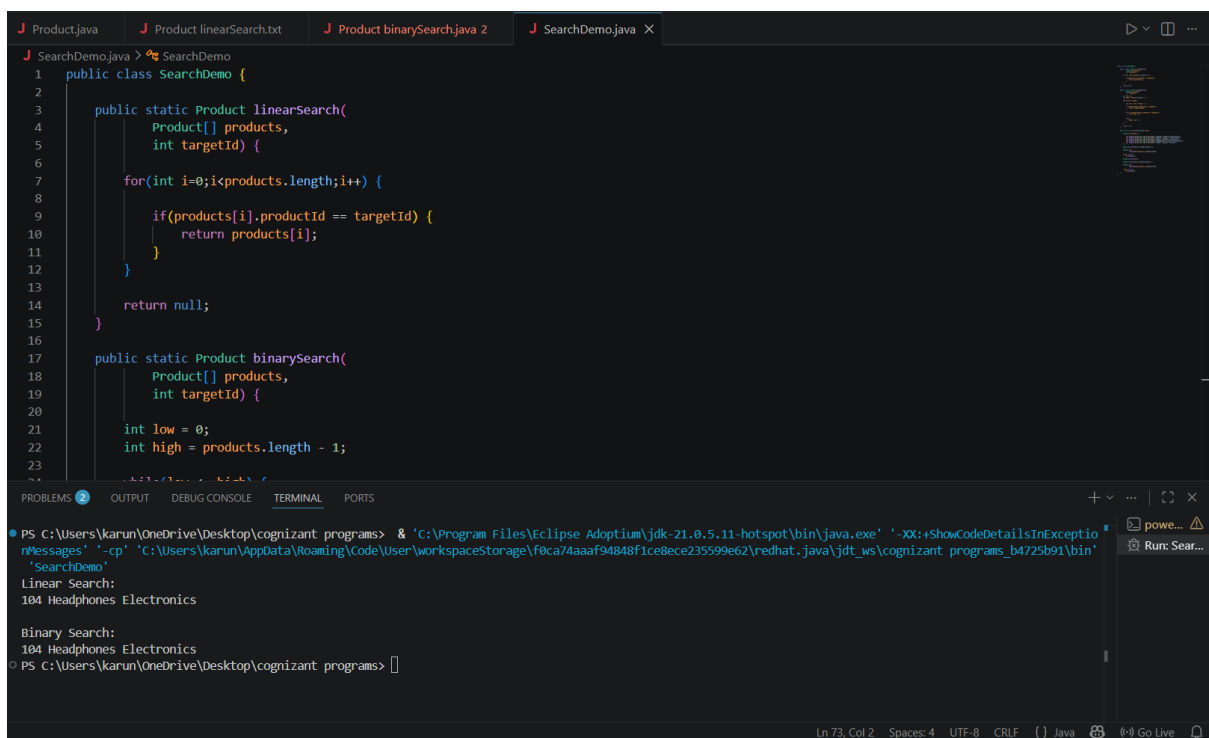


Exercise 2- E-commerce Platform Search Function

The implemented program successfully demonstrates both Linear Search and Binary Search techniques. Linear Search is simple to implement and can be used on both sorted and unsorted datasets. However, its performance decreases as the number of products increases because each element may need to be examined.

Implementation Image



```
SearchDemo.java > SearchDemo
1 public class SearchDemo {
2
3     public static Product linearSearch(
4         Product[] products,
5         int targetId) {
6
7         for(int i=0; i<products.length; i++) {
8
9             if(products[i].productId == targetId) {
10                 return products[i];
11             }
12         }
13
14         return null;
15     }
16
17     public static Product binarySearch(
18         Product[] products,
19         int targetId) {
20
21         int low = 0;
22         int high = products.length - 1;
23
24         // ... (rest of the binary search implementation)
25     }
26 }
```

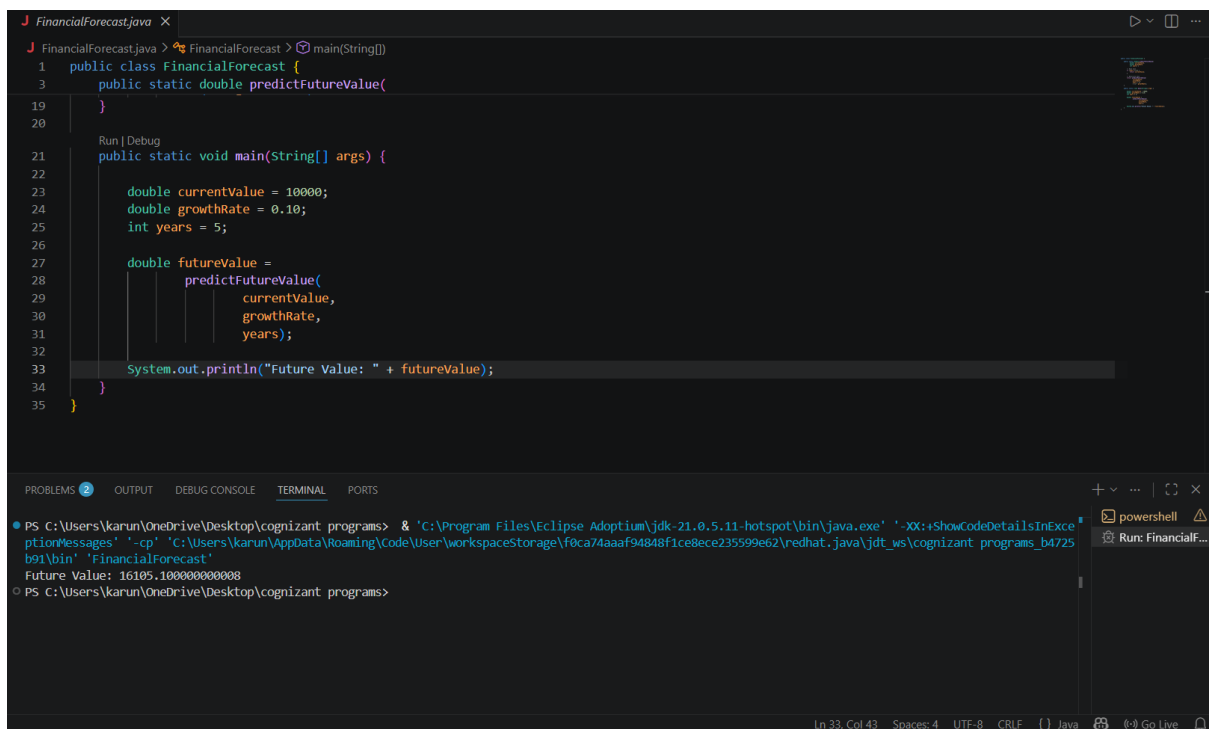
```
PS C:\Users\karun\OneDrive\Desktop\cognizant programs> & 'C:\Program Files\Eclipse Adoptium\jdk-21.0.5.11-hotspot\bin\java.exe' -XX:+ShowCodeDetailsInExceptionMessages -cp 'C:\Users\karun\AppData\Roaming\Code\User\workspaceStorage\f0ca74aaaf94848f1ce8ece235599e62\redhat.java\jdt_ws\cognizant_programs_b4725b91\bin' SearchDemo
Linear Search:
104 Headphones Electronics

Binary Search:
104 Headphones Electronics
PS C:\Users\karun\OneDrive\Desktop\cognizant programs>
```

Exercise 7- Financial Forecasting

The recursive forecasting program successfully predicts future financial values using a fixed annual growth rate. The implementation demonstrates how recursion can model repeated financial growth over multiple periods. The recursive approach closely resembles mathematical formulations and is therefore easy to understand and maintain.

Implementantation Image



```
FinancialForecast.java X
J FinancialForecast.java > FinancialForecast > main(String[])
1 public class FinancialForecast {
3     public static double predictFutureValue(
19     }
20
21 Run | Debug
22 public static void main(String[] args) {
23     double currentValue = 10000;
24     double growthRate = 0.10;
25     int years = 5;
26
27     double futureValue =
28         predictFutureValue(
29             currentValue,
30             growthRate,
31             years);
32
33     System.out.println("Future Value: " + futureValue);
34 }
35 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\karun\OneDrive\Desktop\cognizant programs> & 'C:\Program Files\Eclipse Adoptium\jdk-21.0.5.11-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\karun\AppData\Roaming\Code\User\workspaceStorage\f0ca74aaaf94848f1ce8ece235599e62\redhat.java\jdt_ws\cognizant programs_b4725b91\bin' 'FinancialForecast'

Future Value: 16105.100000000008

PS C:\Users\karun\OneDrive\Desktop\cognizant programs>

Ln 33, Col 43 Spaces: 4 UTF-8 CRLF Java Go Live